Chirality as Curvature: Emergence from Attractor Basin Separatrix Asymmetry

Baoliin (Zaitian) Wu 伍宝林

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1 Physical Proof of Existence

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2 Chirality as Curvature: Emergence from Attractor Basin Separatrix Asymmetry

Baoliin (Zaitian) Wu 伍宝林

ORCiD: 0009-0003-2051-5247

Email: zaitian001@gmail.com

Affiliation: People' s Government of Guangdong Province: Guangzhou, CN

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2.1 Abstract

The origin of homochirality—the predominance of one enantiomeric form in biological systems—remains a fundamental problem in chemistry and the origin of life. Nonlinear effects in asymmetric synthesis and asymmetric autocatalysis demonstrate that small initial chiral imbalances can be amplified under suitable reaction conditions. However, the relationship between reaction kinetics, state-space geometry, and the selection of a particular handedness remains an open theoretical question.

Here we propose a geometric-dynamical architecture in which chiral selection is associated with the geometry of the separatrix between competing enantiomeric attractor basins. We first introduce a small set of **General Postulates of Geometric Selection Theory** that define the mathematical architecture: the statistical state manifold, the mirror involution, the nonequilibrium potential, the separatrix geometry, and the geometric coupling to chiral observables. These postulates function as the architecture-level axioms of the theory, in the same sense that the general postulates of general relativity define its geometric arena.

Within this architecture, we represent accessible chemical states by a statistical manifold equipped with a Fisher information metric and distinguish the metric from the nonequilibrium potential governing chemical dynamics. A consistently defined separatrix-curvature descriptor, denoted by $\Delta\kappa_S$, quantifies a specified form of geometric asymmetry under a defined mirror correspondence.

We establish a **conditional equivalence theorem**: within a model class satisfying explicit structural assumptions, nonzero chiral-selection bias, nonzero separatrix-curvature asymmetry, and noninvariance of the Fisher metric under a mirror involution are equivalent. The proof is mathematically exact within these assumptions; the physical validity of the assumptions for particular chemical systems remains to be established independently.

We further develop two research hypotheses derived from the postulates. First, topological descriptors of chemical state-space dynamics may help characterize the robustness of autocatalytic amplification, provided a nontrivial invariant and its relationship to measurable chemical observables can be established. Second, separatrix geometry may influence nonlinear amplification through its effect on transition actions and rates. A conditional stochastic model illustrates how an exponential dependence of transition rates can produce a hyperbolic-tangent response, while identifying the additional assumptions required for that result.

Four experimentally testable research directions are proposed: reconstructing dynamical state-space geometry, testing topological robustness, measuring geometric contributions to amplification

thresholds, and distinguishing geometry-mediated effects of external fields from conventional chemical mechanisms.

This work is a companion to *Inaugurating the Era of Super Cell: Curvature-Driven Geometric Synthetic Biology for Cell Fate Reprogramming* (DW58553919). Together, the papers propose **Geometric Selection Theory** as a cross-scale research programme connecting molecular handedness and cellular fate selection through the geometry of competing dynamical basins.

Keywords: homochirality; chirality; information geometry; Fisher metric; separatrix; curvature asymmetry; autocatalysis; nonlinear dynamics; topological invariants; Kagan nonlinear effect; Soai reaction; Geometric Selection Theory; postulates

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2.2 1. Introduction

2.2.1 1.1 The Homochirality Problem

Biological systems exhibit a striking preference for particular molecular handedness. Proteins in known terrestrial life predominantly use L-amino acids, while nucleic acids predominantly employ D-sugars. This biochemical regularity raises two distinct questions:

1. How can a chemical system develop a nonzero preference between two enantiomeric states?
2. How can nonlinear reaction networks amplify and maintain such a preference under realistic dynamical and environmental conditions?

These questions must be distinguished. The emergence of a nonzero enantiomeric excess does not by itself explain the magnitude, stability, or ultimate biological universality of homochirality.

In a racemic mixture, the enantiomeric excess is

$$ee = \frac{N_L - N_D}{N_L + N_D}. \quad (1)$$

Here N_L and N_D denote the amounts of the two enantiomers. A racemic mixture has $ee = 0$; a nonzero value indicates an imbalance, while the sign identifies which enantiomer predominates under the stated convention.

The challenge is to understand how reaction dynamics can transform a small initial imbalance into a substantial final excess, and under what conditions such amplification occurs.

2.2.2 1.2 Nonlinear Amplification and Asymmetric Autocatalysis

Research on nonlinear effects in asymmetric synthesis and the Soai reaction established important experimental examples of chiral amplification.

The nonlinear effect investigated by Puchot and colleagues showed that product enantiomeric excess need not depend linearly on catalyst enantiomeric excess. The relationship depends on the reaction mechanism, catalyst aggregation, and other chemical conditions.

The Soai reaction provides a particularly important example of asymmetric autocatalysis: a chiral product participates in catalysing its own formation, allowing a small chiral imbalance to be amplified under suitable conditions.

These findings motivate a geometric question:

Can the geometry of the accessible chemical state space provide additional predictive information about chiral selection beyond the reaction-rate parameters traditionally used to describe it?

This question does not presume that kinetic explanations are inadequate. Rather, it asks whether geometric quantities can be derived from, or add predictive power to, established dynamical models.

2.2.3 1.3 Existing Dynamical Explanations

Frank' s model describes spontaneous asymmetric synthesis through autocatalytic production and mutual inhibition between enantiomeric species. Subsequent stochastic and reaction-network models have explored symmetry-breaking transitions, fluctuations, bifurcations, and nonlinear amplification.

These models establish an important point: chiral selection can arise from the structure and dynamics of a chemical reaction network without requiring a pre-existing large chiral imbalance.

Their mathematical descriptions also provide a natural starting point for the present work. Nonlinear reaction systems can possess multiple stable states, unstable intermediate states, and basin boundaries separating competing outcomes.

The present proposal is to investigate whether geometric descriptors of these structures can help characterize the transition between racemic and chiral states.

2.2.4 1.4 The Geometric-Dynamical Hypothesis

We propose that chiral selection may be associated with the geometry of the separatrix between competing enantiomeric attractor basins.

The hypothesis has four components.

First, statistical state geometry. Accessible chemical states are represented by a statistical manifold $\mathcal{M}$ with a Fisher information metric g, where such a metric is well defined and non-degenerate.

Second, dynamical basin structure. A specified reaction model defines the dynamical evolution of chemical states. Where competing attractors exist, their basins are separated by a boundary or separatrix.

Third, geometric asymmetry. A carefully defined descriptor $\Delta\kappa_{\mathcal{S}}$ quantifies a specified asymmetry of the separatrix under a mirror correspondence.

Fourth, nonlinear state selection. A dynamical coupling between this geometric descriptor and a measurable chiral-selection bias is proposed as a model-specific hypothesis.

The fourth component is essential. Geometry alone does not establish that one enantiomer will be selected. A quantitative relationship between geometry and dynamics must be derived or experimentally supported.

2.2.5 1.5 Relationship to the Super Cell Paper

This manuscript is a companion to *Inaugurating the Era of Super Cell: Curvature-Driven Geometric Synthetic Biology for Cell Fate Reprogramming* (DW58553919).

Concept	Molecular scale	Cellular scale
State space	Chemical state manifold	Cell-fate state manifold
Competing states	Enantiomeric states	Cellular phenotypes
Dynamical structure	Chemical attractor basins	Phenotypic attractor basins
Basin boundary	Chemical separatrix	Cell-fate separatrix
Geometric descriptor	Separatrix-curvature asymmetry	Corresponding geometric descriptor
Selection process	Chiral amplification	Cell-fate selection or reprogramming
Central research question	Does geometry improve prediction of chiral selection?	Does geometry improve prediction of cell-fate transitions?

The two works share a research architecture but concern different physical systems. Their common structure does not establish that the same constitutive equations or quantitative laws apply at both scales.

We use **Geometric Selection Theory** to denote this proposed research programme: the study of whether geometric and topological properties of competing dynamical basins can provide predictive principles for discrete state selection in complex systems.

2.2.6 1.6 Structure of the Paper

Section 2 introduces the **General Postulates of Geometric Selection Theory**, followed by the statistical and dynamical state-space architecture. Section 3 defines separatrix-curvature asymmetry and proves the conditional equivalence theorem. Section 4 examines the possible role of topology in autocatalytic robustness. Section 5 develops a conditional stochastic model of nonlinear amplification. Section 6 presents four testable research directions. Section 7 discusses the implications, limitations, and relationship to the companion paper. Section 8 concludes.

2.3 2. General Postulates and Theoretical Architecture

2.3.1 2.1 General Postulates of Geometric Selection Theory

Before developing any specific model, we state a small set of general postulates that define the mathematical architecture of Geometric Selection Theory. These postulates play the same role as the general postulates of general relativity: they define the geometric arena and the correspondence rules, but they do not by themselves predict any specific numerical outcome. They are accepted as the architecture' s starting points and are not individually falsifiable; the specific hypotheses and predictions derived from them are.

Postulate	Mathematical Formulation	Role in Architecture
P1. Statistical State Manifold	Accessible states of the system form a smooth statistical manifold $\mathcal{M}$ with a positive-definite Fisher information metric g_{ab} where the statistical model is regular and identifiable.	Defines the geometric arena.
P2. Mirror Involution	The exchange of enantiomeric or competing states corresponds to a smooth involution $\mathcal{P} : \mathcal{M} \to \mathcal{M}$ with $\mathcal{P}^2 = \mathrm{id}$.	Defines the symmetry operation.
P3. Nonequilibrium Potential	Under appropriate conditions, the dynamics admit a quasipotential Φ whose covariant Hessian $H_{ab} = \nabla_a \nabla_b \Phi$ governs the stability structure, alongside non-conservative components.	Defines the dynamical driving.
P4. Separatrix Geometry	The boundary $\mathcal{S}$ between competing attractor basins is a sufficiently regular hypersurface whose geometry can be characterized by curvature invariants.	Defines the geometric descriptor.
P5. Geometric Coupling to Chiral Observables	Within the model class, the chiral-selection observable B couples to the separatrix-curvature asymmetry $\Delta\kappa_{\mathcal{S}}$ through a nonzero coefficient.	Bridges geometry and chemistry.

Remarks on the logical role of the postulates.

- P1–P4 define the geometric and dynamical structure of the architecture. They are architecture-level commitments, not empirical claims about any particular reaction.
- P5 is the bridge postulate that connects the geometric descriptor to a measurable chemical quantity. It is the most physically substantive of the five and the one most likely to require model-specific justification in applications.
- The postulates do not assert universal numerical laws. They assert that the architecture' s objects exist and are related in the specified way.
- All specific predictions in Sections 3–6 are conditional on the postulates and on additional model-specific assumptions.

2.3.2 2.2 Statistical Representation of Chemical States

Let $\mathcal{M}$ denote a parameter manifold describing the accessible macroscopic states of a chemical system.

A state $\theta \in \mathcal{M}$ is associated with a probability distribution over microscopic configurations:

$$p(x \mid \theta). \quad (2)$$

The coordinates $\theta = (\theta^1, \ldots, \theta^n)$ may include concentrations, enantiomeric excess, temperature, and other observables, provided the chosen parametrization defines a statistically identifiable model.

We adopt the following modelling premises, which refine Postulate P1.

Assumption A1: Statistical state representation. The chemical states under investigation admit a differentiable family of probability distributions with identifiable parameters.

Assumption A2: Specified statistical ensemble. Where a maximum-entropy representation is used, the relevant constraints and statistical ensemble are explicitly specified. In an exponential-family model,

$$p(x \mid \lambda) = \exp\left[\sum_a \lambda^a f_a(x) - \psi(\lambda)\right], \quad (3)$$

where $\psi(\lambda)$ is the log-partition function and the distribution is normalized.

Assumption A3: Statistical consistency. The inference procedure respects the relevant invariance properties under admissible reparametrization and sufficient-statistic transformations.

These assumptions establish a modelling architecture; they do not imply that every chemical system is in equilibrium, maximizes entropy under the same constraints, or possesses a globally identifiable statistical manifold.

2.3.3 2.3 Fisher Information Geometry

For a differentiable statistical model, the Fisher information metric is defined by

$$g_{ab}(\theta) = \mathbb{E}_\theta\left[\frac{\partial \ln p}{\partial \theta^a}\frac{\partial \ln p}{\partial \theta^b}\right]. \quad (4)$$

When the required regularity conditions hold, this metric measures the local distinguishability of nearby probability distributions.

For the exponential-family representation in Equation (3), the Fisher metric in natural-parameter coordinates satisfies

$$g_{ab}(\lambda) = \frac{\partial^2 \psi}{\partial \lambda^a \partial \lambda^b} = \mathrm{Cov}_\lambda(f_a, f_b). \quad (5)$$

If $\eta_a = \mathbb{E}_\lambda[f_a]$ denotes the corresponding expectation parameter and the relevant Legendre transformation is regular, then the dual coordinate representation satisfies

$$g_{ab}^{(\eta)} = -\frac{\partial^2 S}{\partial \eta_a \partial \eta_b}, \quad (6)$$

where S is the Shannon entropy of the corresponding exponential-family distribution.

Equations (5) and (6) express dual-coordinate representations of information geometry. They must not be interpreted as asserting that the same Hessian matrix represents the metric in arbitrary coordinates.

Under the regularity conditions of Chentsov' s theorem, the Fisher metric is characterized, up to an overall positive factor, by its invariance under appropriate stochastic transformations and sufficient-statistic operations.

Proposition 1. For a regular statistical model satisfying the conditions of the relevant information-geometric uniqueness theorem, the Fisher information metric provides the canonical metric associated with statistical distinguishability, up to the permitted normalization.

The proposition does not establish a universal physical law for chemical dynamics. It supplies a principled metric for describing the statistical state space when the assumptions are satisfied.

2.3.4 2.4 Nonequilibrium Dynamics and the Potential Landscape

Chemical amplification generally involves nonequilibrium processes. We therefore distinguish the statistical metric g from the function governing dynamical evolution.

Let $\Phi : \mathcal{M} \to \mathbb{R}$ denote a specified dynamical potential or, where appropriate, a nonequilibrium quasipotential.

Its covariant Hessian is

$$H_{ab} = \nabla_a \nabla_b \Phi. \quad (7)$$

The metric g and the Hessian H are distinct objects. The Fisher metric is positive definite when the statistical model is regular and identifiable. The Hessian of a nonequilibrium potential may be indefinite, particularly near saddle points and basin boundaries.

A quasipotential is not automatically a thermodynamic free energy. Its meaning depends on the stochastic dynamical model from which it is derived.

Where the dynamics admit two stable enantiomeric states, we denote their attractors by A_L and A_D, and their basins by B_L and B_D. The boundary between these basins is denoted by

$$\mathcal{S} = \partial B_L \cap \partial B_D, \quad (8)$$

when this expression accurately describes the relevant basin boundary.

The existence and regularity of $\mathcal{S}$ must be established for the chosen dynamical system. In gradient systems, basin boundaries may be associated with stable manifolds of saddle points, but this relationship is not universal for arbitrary nonequilibrium dynamics.

2.3.5 2.5 Mirror Involution and Nonlinear Symmetry Breaking

Let

$$\mathcal{P} : \mathcal{M} \to \mathcal{M}, \qquad \mathcal{P}^2 = \mathrm{id}, \quad (9)$$

denote a smooth involution representing the exchange of the two enantiomeric states.

For a mirror-symmetric statistical model, the metric satisfies

$$\mathcal{P}^* g = g, \quad (10)$$

where $\mathcal{P}^* g$ is the pullback metric.

For the dynamics to be mirror-equivariant, the dynamical law must also transform consistently under $\mathcal{P}$. Metric invariance alone does not guarantee that the reaction dynamics, external fields, boundary conditions, or nonequilibrium driving are mirror-symmetric.

We therefore distinguish:

- **Mirror symmetry:** invariance under a specified mirror involution.
- **Nonlinear dynamical symmetry:** invariance or equivariance of a specified nonlinear evolution law under a defined transformation.
- **Mirror-symmetry breaking:** failure of the relevant mirror-invariance condition in the specified system or state.
- **Nonlinear chiral selection:** dynamical amplification or selection of one enantiomeric state over the other.

This terminology preserves mathematical precision while emphasizing the nonlinear dynamics central to the proposed theory.

2.4 3. Chirality as Separatrix-Curvature Asymmetry

2.4.1 3.1 Defining the Geometric Descriptor

The central geometric object is the separatrix $\mathcal{S}$ between competing attractor basins.

In a two-dimensional Riemannian state manifold, a sufficiently regular separatrix is locally a curve. Let

$$\gamma(s) \in \mathcal{S} \quad (11)$$

be a unit-speed parametrization, where s is arc length.

Its geodesic curvature is

$$\kappa_g(s) = \left\langle \nabla_{\dot\gamma} \dot\gamma, n \right\rangle_g, \quad (12)$$

where n is a specified unit normal to the curve within the state manifold.

In higher-dimensional manifolds, the separatrix is generally a hypersurface. Its geometry may be characterized by the second fundamental form, principal curvatures, mean curvature, or another explicitly defined geometric invariant.

A signed curvature depends on the chosen normal orientation. Consequently, comparing curvature at mirror-related points requires a consistent convention for the mirror map, tangent correspondence, and normal transformation.

Let s and $\mathcal{P}(s)$ denote corresponding points under the prescribed mirror operation. Define the separatrix-curvature asymmetry descriptor by

$$\Delta\kappa_{\mathcal{S}}(s) = \kappa_{\mathcal{S}}(s) - \kappa_{\mathcal{S}}^{\mathrm{mir}}(s), \quad (13)$$

where $\kappa_{\mathcal{S}}^{\mathrm{mir}}(s)$ is the curvature predicted at the corresponding point under the explicitly specified mirror transformation and orientation convention.

Equation (13) is an operational definition, not a universal scalar invariant. Its coordinate independence and invariance under admissible reparametrizations must be demonstrated for the particular construction.

For a global analysis, one may define a scalar descriptor such as

$$\mathcal{K}_{\mathcal{S}} = \int_{\mathcal{S}} w(s)\, \Delta\kappa_{\mathcal{S}}(s)\, d\mu_{\mathcal{S}}, \quad (14)$$

where $w(s)$ is a specified weighting function and $d\mu_{\mathcal{S}}$ is the induced measure on the separatrix.

The weighting, normalization, orientation convention, and physical units must be specified before comparing different systems. If the signed contributions cancel, a zero global integral does not necessarily imply pointwise mirror symmetry. Accordingly, the descriptor used in the theorem must be chosen to have the required diagnostic properties.

For notational simplicity, the remainder of this section writes the selected descriptor as

$$\Delta\kappa_{\mathcal{S}}. \quad (15)$$

2.4.2 3.2 Chiral Bias as an Observable

We define a signed chiral-selection bias B through an explicitly specified observable.

For a system whose relevant observable is the enantiomeric excess, one possible choice is

$$B = ee_{\mathrm{ss}}, \quad (16)$$

where ee_{ss} denotes the enantiomeric excess in a specified steady state or other explicitly defined observation window.

This choice is appropriate only when such a state or observation window is well defined. For time-dependent reactions, one may instead define B as the excess at a prescribed time or as a specified time-averaged quantity.

The distinction matters because a system can exhibit a nonzero transient excess while ultimately returning to a racemic state, or can display unequal transition rates without having unequal equilibrium free energies.

In this manuscript, $B \neq 0$ means that the selected observable exhibits a nonzero preference under the stated measurement protocol.

2.4.3 3.3 The Central Theorem

Theorem 2 (Conditional Equivalence of Chiral Bias, Separatrix-Curvature Asymmetry, and Metric Symmetry Breaking).

Let $(\mathcal{M}, g)$ be a smooth statistical state manifold describing a chemical system with two competing states related by a smooth involution $\mathcal{P}$, where $\mathcal{P}^2 = \mathrm{id}$.

Let $\mathcal{S}$ be a sufficiently regular separatrix between the corresponding dynamical basins. Let $\Delta\kappa_{\mathcal{S}}$ be a consistently defined scalar descriptor of separatrix-curvature asymmetry.

Suppose the following conditions hold.

(C1) Regular statistical geometry. The state manifold admits a differentiable, non-degenerate Fisher information metric g, and the mirror involution acts smoothly on the manifold.

(C2) Identifiable separatrix geometry. The separatrix and its curvature descriptor are well defined, and $\Delta\kappa_{\mathcal{S}}$ is invariant under the admissible coordinate transformations and orientation conventions specified by the model.

(C3) Faithful geometric order parameter. Within the selected model class, the curvature descriptor detects precisely the relevant mirror-symmetry breaking of the metric:

$$\Delta\kappa_{\mathcal{S}} = 0 \quad \Longleftrightarrow \quad \mathcal{P}^* g = g. \quad (17)$$

This is a model-specific structural condition. It does not follow from the existence or uniqueness of the Fisher metric alone.

(C4) Nontrivial geometric coupling to selection. The chiral-selection observable satisfies

$$B = \lambda \Delta\kappa_{\mathcal{S}}, \qquad \lambda \neq 0, \quad (18)$$

where λ is a finite coupling coefficient with the units required by the chosen normalization.

Equation (18) is a constitutive hypothesis, corresponding to Postulate P5 specialized to the model class. It must be derived from a specified dynamical model or supported by independent quantitative evidence.

Then, within this model class, the following statements are equivalent:

(i) The selected observable exhibits nonzero chiral bias, $B \neq 0$.

(ii) The separatrix-curvature asymmetry is nonzero, $\Delta\kappa_{\mathcal{S}} \neq 0$.

(iii) The Fisher metric is not invariant under the mirror involution, $\mathcal{P}^* g \neq g$.

Proof.

By Assumption (C4),

$$B = \lambda \Delta\kappa_{\mathcal{S}}, \qquad \lambda \neq 0. \quad (19)$$

Since λ is nonzero,

$$B \neq 0 \quad \Longleftrightarrow \quad \Delta\kappa_{\mathcal{S}} \neq 0. \quad (20)$$

Thus, statements (i) and (ii) are equivalent.

By Assumption (C3),

$$\Delta\kappa_{\mathcal{S}} = 0 \quad \Longleftrightarrow \quad \mathcal{P}^* g = g. \quad (21)$$

Taking the logical negation of both sides gives

$$\Delta\kappa_{\mathcal{S}} \neq 0 \quad \Longleftrightarrow \quad \mathcal{P}^* g \neq g. \quad (22)$$

Thus, statements (ii) and (iii) are equivalent.

Combining Equations (20) and (22) yields

$$\boxed{B \neq 0 \quad \Longleftrightarrow \quad \Delta\kappa_{\mathcal{S}} \neq 0 \quad \Longleftrightarrow \quad \mathcal{P}^* g \neq g}. \quad (23)$$

Therefore, the three statements are equivalent under Assumptions (C1)-(C4). □

2.4.4 3.4 Interpretation and Scope

Theorem 2 provides a mathematically explicit conditional relationship between chiral bias, separatrix geometry, and metric symmetry breaking.

Its logical structure is important.

First, the equivalence between $B \neq 0$ and $\Delta\kappa_{\mathcal{S}} \neq 0$ follows from the nonzero coupling in C4.

Second, the equivalence between $\Delta\kappa_{\mathcal{S}} \neq 0$ and $\mathcal{P}^* g \neq g$ follows from C3.

The proof does not independently establish either structural condition. In particular, a scalar curvature descriptor does not generally determine an entire metric, and metric asymmetry need not be detectable by every possible scalar descriptor.

Likewise, a nonzero geometric descriptor does not automatically cause a chemical system to select one enantiomer. The causal or constitutive relationship must be established for the specified dynamics.

The theorem should therefore be interpreted as a conditional equivalence theorem within a restricted model class, not as an unconditional law of stereochemistry.

2.4.5 3.5 Physical Interpretation

The proposed physical interpretation is that separatrix geometry may encode information about the relative accessibility of competing dynamical outcomes.

A curvature descriptor alone, however, does not establish that one basin is energetically deeper, that a barrier is lower, or that a transition rate is greater. Those conclusions require a specified relationship between geometry and the dynamical law.

A geometric contribution to selection might arise if the geometry of the state manifold affects the optimal transition path, the local noise structure, the effective action, or another quantity controlling basin-to-basin transitions.

The principal research question is therefore not simply whether $\Delta\kappa_{\mathcal{S}} \neq 0$, but whether this descriptor improves quantitative predictions of chiral selection beyond established kinetic and thermodynamic variables.

2.5 4. Autocatalysis, Dynamical Cycles, and Topological Robustness

2.5.1 4.1 Autocatalysis as a Dynamical Process

An autocatalytic reaction is characterized by a product or product-containing species participating in the production of additional material of the same catalytic type.

The resulting chemical dynamics may include feedback, transient oscillations, stable fixed points, limit cycles, or more complicated trajectories, depending on the reaction network and operating conditions.

Let a chemical state evolve according to

$$\dot{x} = F(x; \mu), \quad (24)$$

where x represents the chemical state variables and μ denotes the kinetic parameters and external controls.

A periodic solution satisfies

$$x(t + T) = x(t), \qquad T > 0. \quad (25)$$

Periodicity does not imply that the trajectory is a geodesic of the Fisher metric. A geodesic instead satisfies

$$\nabla_{\dot{\gamma}} \dot{\gamma} = 0. \quad (26)$$

To identify a chemical cycle with a geodesic, one would need to derive the relevant metric, parameterize the trajectory appropriately, and demonstrate that the geodesic equation holds. No such identification is assumed universally here.

The geometric proposal is more general: dynamical cycles and their state-space structures may provide useful descriptors of autocatalytic organization, whether or not the cycles are geodesics.

2.5.2 4.2 Topological Invariants and Their Requirements

A closed trajectory γ defines a homology class

$$[\gamma] \in H_1(\mathcal{M}, \mathbb{Z}) \quad (27)$$

when the relevant topological construction is well defined.

If $[\gamma]$ is nontrivial, continuous deformations within the applicable class may preserve that homology class. This fact alone does not guarantee that the periodic orbit persists under every perturbation, nor does it establish any conserved chemical observable.

Similarly, a Berry connection $\mathcal{A}$ may define a geometric phase along a closed path,

$$\varphi_{\mathrm{B}} = \oint_{\gamma} \mathcal{A}, \quad (28)$$

but this phase is not generally an integer multiple of 2π. Quantization requires additional structure and conditions.

A genuine topological protection claim must therefore identify:

1. The relevant topological space or bundle.
2. The invariant being calculated.
3. The class of perturbations under which the invariant is preserved.
4. The mathematical result establishing persistence or robustness.
5. The independently measurable chemical quantity linked to the invariant.

Without these elements, topological protection remains a research hypothesis derived from Postulate P4 rather than an established mechanism.

2.5.3 4.3 Hypothesis T1: Topological Contribution to Autocatalytic Robustness

Hypothesis T1. For a specified autocatalytic reaction network, a suitably defined topological descriptor of its dynamical state space may contribute predictive information about the persistence of dynamical regimes under continuous perturbations.

This hypothesis is derived from Postulate P4 (separatrix geometry) and the topological structure of the state manifold. It does not assert that the enantiomeric excess is itself a topological invariant, nor that it cannot change continuously.

A candidate mathematical programme is:

- Construct a dynamical model from a specified reaction network.
- Identify stable states, periodic orbits, basin boundaries, and relevant invariant sets.
- Compute appropriate topological descriptors where mathematically justified.
- Determine how these descriptors change under controlled perturbations.
- Compare their predictive performance with conventional kinetic models.

The topological contribution would be supported only if a reproducible relationship between the invariant and a measurable property of the reaction were demonstrated.

2.5.4 4.4 Application to the Soai Reaction

The Soai reaction is a useful test case because it exhibits strong asymmetric autocatalytic amplification.

The proposed geometric-topological analysis would ask whether a state-space representation of the reaction provides additional predictive information about:

- the dependence of product ee on initial ee;
- the stability of observed amplification under perturbations;
- the existence and location of dynamical transitions;
- the role of aggregation and catalyst speciation;
- the relationship between basin structure and the probability of selecting either enantiomer.

These questions must be studied using a chemically appropriate kinetic model. A geometric descriptor should not replace established reaction mechanisms without demonstrating equivalent or superior explanatory and predictive performance.

2.5.5 4.5 Distinguishing Robustness from Conservation

A crucial distinction is that robustness does not necessarily mean invariance.

A reaction may remain strongly biased toward one enantiomer across a broad range of conditions while its measured ee changes continuously. This is compatible with dynamical robustness but does not imply topological conservation of ee.

Accordingly, the present work proposes topology as a possible source of structural robustness in selected dynamical regimes. It does not claim that the chiral purity of the Soai reaction is protected against all continuous perturbations by a winding number.

2.6 5. Nonlinear Amplification as a Conditional Geometric Effect

2.6.1 5.1 The General Nonlinear Response

Let e_c denote catalyst enantiomeric excess and e_p denote product enantiomeric excess under a specified reaction protocol.

The general relationship is

$$e_p = f(e_c; \mu), \quad (29)$$

where μ denotes other relevant reaction parameters.

The function f depends on the reaction mechanism. Nonlinear behavior may result from autocatalysis, catalyst aggregation, mutual inhibition, saturation, competing pathways, or other kinetic processes.

The geometric hypothesis asks whether a separatrix descriptor can explain or predict part of this response.

2.6.2 5.2 A Conditional Stochastic Model

Consider a stochastic dynamical system whose rare transitions between two metastable states are described by a large-deviation approximation.

Under appropriate assumptions, the transition rate from state i to state j can take the form

$$\Gamma_{i\to j} \asymp A_{ij} \exp\left(-\frac{S_{i\to j}}{\epsilon_n}\right), \quad (30)$$

where $S_{i\to j}$ is the relevant minimum action, A_{ij} is a prefactor, and ϵ_n is a small-noise scale.

The symbol $\asymp$ indicates leading exponential-order behavior rather than an exact equality.

Suppose, as a model-specific hypothesis derived from Postulate P5, that a small geometric perturbation changes one transition action according to

$$S_{L\to D} = S_0 - a\Delta\kappa_{\mathcal{S}} + O((\Delta\kappa_{\mathcal{S}})^2), \quad (31)$$

where a is a coupling coefficient determined by the dynamics and the normalization of the curvature descriptor.

Then, at leading exponential order,

$$\Gamma_{L\to D} \asymp A_{L\to D} \exp\left[-\frac{S_0 - a\Delta\kappa_{\mathcal{S}}}{\epsilon_n}\right]. \quad (32)$$

If the prefactor varies slowly and higher-order terms are negligible, this expression suggests an exponential sensitivity of the transition rate to the geometric perturbation.

Equation (31) is not a universal identity. It must be derived for the particular stochastic system. In particular, the coefficient a depends on the action functional, drift, noise covariance, and the way in which the geometric perturbation enters the model.

2.6.3 5.3 A Hyperbolic-Tangent Response Under Additional Assumptions

A hyperbolic-tangent relationship follows if the two transition rates satisfy the specific symmetric exponential form

$$\Gamma_{L\to D} = \Gamma_0 e^{x}, \qquad \Gamma_{D\to L} = \Gamma_0 e^{-x}, \quad (33)$$

where

$$x = \frac{a\Delta\kappa_{\mathcal{S}}}{\epsilon_n}. \quad (34)$$

The normalized rate imbalance is then

$$B_{\text{rate}} = \frac{\Gamma_{L\to D} - \Gamma_{D\to L}}{\Gamma_{L\to D} + \Gamma_{D\to L}}. \quad (35)$$

Substituting Equation (33) gives

$$B_{\text{rate}} = \frac{e^{x} - e^{-x}}{e^{x} + e^{-x}} = \tanh x. \quad (36)$$

Thus,

$$\boxed{B_{\text{rate}} = \tanh\left(\frac{a\Delta\kappa_{\mathcal{S}}}{\epsilon_n}\right)}. \quad (37)$$

For small x,

$$B_{\text{rate}} = x + O(x^3), \quad (38)$$

whereas for large positive or negative x, the rate imbalance approaches $+1$ or -1, respectively.

This provides a mathematically explicit example of a nonlinear response generated by exponential transition-rate sensitivity.

However, three qualifications are essential.

First, Equation (37) describes a normalized transition-rate imbalance. It does not automatically equal the final product ee.

Second, the symmetric rate form in Equation (33) is an additional assumption. It must be justified by the particular dynamical model.

Third, even if the rate imbalance follows Equation (37), the mapping from this quantity to the final chemical product distribution depends on the reaction network and observation protocol.

Accordingly, Equation (37) is a conditional model result, not a universal derivation of the Kagan nonlinear effect.

2.6.4 5.4 Hypothesis G1: Geometric Contribution to Nonlinear Amplification

Hypothesis G1. Within a specified stochastic reaction model, separatrix geometry may influence nonlinear chiral amplification through its effect on transition actions, rates, or basin accessibility.

This hypothesis is derived from Postulate P5 applied to the specific transition-rate structure of the model.

The model should define an operational amplification threshold. For example, one may define I_{th} as the smallest value of a specified control parameter I at which a pre-registered amplification criterion is met.

A possible dimensionless scaling hypothesis is

$$I_{\text{th}} = F\left(\widehat{\mathcal{K}}_{\mathcal{S}}, \hat{\mu}, \hat{\epsilon}_n\right), \quad (39)$$

where the hatted variables denote dimensionless, explicitly defined quantities.

A linear relation between I_{th} and a curvature descriptor may be tested as a special case, but it cannot be assumed without a model-specific derivation.

This formulation avoids assigning a universal proportionality between a dimensional curvature and an unspecified threshold. It also permits a fair comparison with conventional kinetic explanations.

2.6.5 5.5 What Would Establish a Geometric Contribution?

A convincing result would require more than an observed correlation between a fitted curvature descriptor and product ee.

At minimum, the analysis should:

1. Specify the reaction network and kinetic equations.
2. Define the state manifold and curvature descriptor independently of the final ee.
3. Derive how the descriptor enters the dynamical equations or transition action.
4. Estimate parameters using a calibration dataset.
5. Compare geometric and conventional kinetic models on independent data.
6. Quantify uncertainty and test whether the geometric descriptor improves prediction.

Only after these steps could the geometric mechanism be considered quantitatively supported for the reaction under investigation.

2.7 6. Experimental Predictions and Falsifiable Tests

The following research directions are designed to distinguish the geometric hypothesis from established explanations. Each prediction requires operational definitions and independent measurements.

2.7.1 6.1 Prediction 1: Geometric Descriptors Improve Prediction of Chiral Selection

Prediction. A properly reconstructed state-space geometry may provide predictive information about chiral selection beyond the variables included in an established kinetic model.

For a specified reaction network, let X denote the conventional kinetic variables and let $\mathcal{K}_{\mathcal{S}}$ denote an independently estimated geometric descriptor.

The test is whether

$$P(ee \mid X, \mathcal{K}_{\mathcal{S}}) \quad (40)$$

predicts held-out measurements more accurately than

$$P(ee \mid X) \quad (41)$$

under a pre-specified evaluation protocol.

Experimental and computational protocol:

1. Select a reaction with reproducible chiral amplification.

2. Construct a kinetic model and identify its accessible dynamical states.
3. Reconstruct the relevant state-space geometry using independently defined observables.
4. Estimate the geometric descriptor without using the final ee as the sole fitting target.
5. Compare predictions on independent reaction conditions.
6. Quantify uncertainty and assess sensitivity to the reconstruction procedure.

The prediction would be weakened if the geometric descriptor failed to improve out-of-sample performance or if its apparent contribution disappeared after controlling for established kinetic variables.

This prediction does not assert that isolated mirror-image molecules must possess different energy Hessian spectra. Under mirror-symmetric conditions, corresponding enantiomers are expected to have equivalent energy spectra after the appropriate coordinate transformation.

2.7.2 6.2 Prediction 2: Topological Descriptors May Predict Dynamical Robustness

Prediction. For a specified autocatalytic network, an independently defined topological invariant may help predict the persistence of a dynamical regime under a defined class of perturbations.

Experimental and computational protocol:

1. Specify the reaction network and dynamical model.
2. Identify the invariant set or periodic orbit to be studied.
3. Define the topological invariant and the admissible perturbation class.
4. Establish the mathematical conditions under which the invariant is preserved.
5. Measure dynamical stability and chiral amplification across controlled perturbations.
6. Compare the invariant' s predictive contribution with conventional stability measures.

A plateau in product ee would not, by itself, establish topological protection. Evidence would require a demonstrated relationship between the invariant and the persistence of a defined dynamical property, together with appropriate controls.

The hypothesis would be weakened if the proposed invariant were undefined, changed under the stated perturbations, or failed to predict robustness beyond conventional dynamical variables.

2.7.3 6.3 Prediction 3: Geometric Contributions to Amplification Thresholds

Prediction. If separatrix geometry affects transition actions, an appropriately normalized geometric descriptor may correlate with the onset or magnitude of nonlinear amplification across a controlled catalyst series.

Protocol:

1. Select a chemically related series of catalysts or reaction conditions.
2. Measure catalyst ee, product ee, conversion, reaction rates, and relevant speciation variables.
3. Reconstruct or calculate the geometric descriptor using a pre-specified procedure.
4. Define the amplification threshold before examining the comparative results.
5. Fit a model relating the threshold to geometric and conventional kinetic variables.
6. Evaluate the model on independent data.

A proposed linear scaling should be tested as one candidate model rather than assumed in advance. Competing nonlinear, saturating, and kinetic-only models should be included.

The prediction would be supported if the geometric descriptor exhibited reproducible explanatory or predictive value, with uncertainty estimates that excluded trivial or confounded explanations.

2.7.4 6.4 Prediction 4: External Perturbations and Geometry-Mediated Chiral Selection

Prediction. An external field may alter chiral reaction outcomes through changes in the reaction dynamics and potentially through changes in a reconstructed state-space geometric descriptor.

This possibility is termed **geometry-mediated chiral selection** in the present architecture. It is not assumed that an external field automatically creates a geometric autocatalytic mechanism.

Protocol:

1. Select a reaction whose response to an external field can be measured reproducibly.
2. Apply controlled field strengths and orientations.
3. Measure ee, conversion, reaction rates, and other relevant chemical observables.
4. Include field-free controls and controls for heating, transport, photochemistry, and other known field-mediated mechanisms.
5. Independently estimate the proposed geometric descriptor.
6. Test whether changes in the descriptor predict reaction outcomes beyond established physical and chemical variables.

An observed field-induced ee would not alone demonstrate a geometric mechanism. The hypothesis requires evidence that the geometric descriptor has an independent, reproducible relationship to the response.

2.7.5 6.5 Summary of Predictions

Prediction	Measurable target	Principal test
Geometric contribution to chiral selection	Held-out prediction of ee	Compare kinetic-only and geometry-augmented models
Topological contribution to robustness	Persistence of a defined dynamical regime	Test invariant-based predictions under controlled perturbations
Geometric contribution to thresholds	Amplification onset or magnitude	Compare dimensionless threshold models across a catalyst series
Geometry-mediated external-field response	Field-dependent ee and geometric descriptors	Separate geometric contributions from conventional field effects

All four predictions are falsifiable. None should be treated as established merely because the proposed geometric architecture can accommodate the corresponding observation.

2.8 7. Discussion

2.8.1 7.1 The Main Mathematical Contribution

The principal mathematical contribution is Theorem 2, which establishes a conditional equivalence between a defined chiral-selection observable, a specified separatrix-curvature descriptor, and

mirror-symmetry breaking of the Fisher metric.

The proof is straightforward once the structural assumptions are accepted. The substantive scientific question is whether those assumptions can be derived or validated for chemically realistic systems.

The theorem therefore provides a precise research target: construct a model in which the curvature descriptor is well defined, demonstrate that it faithfully detects the relevant metric symmetry breaking, and independently establish its coupling to chiral selection.

This approach is more restrictive than a general claim that geometry influences chemistry, but it is also more testable.

2.8.2 7.2 The Role of the Postulates

The General Postulates of Geometric Selection Theory (Section 2.1) define the mathematical architecture in the same sense that the general postulates of general relativity define the geometric arena of gravitational physics. They are not empirical claims about any particular reaction and are not individually falsifiable.

The scientific content of the theory resides in:

- the conditional equivalence theorem (Theorem 2), which is a rigorous mathematical result within the architecture;
- the research hypotheses T1 and G1, which are derived from the postulates and are testable;
- the four experimental predictions, which are designed to distinguish the geometric hypothesis from established explanations.

This three-level structure—postulates, theorems, hypotheses—preserves both the coherence of the architecture and the falsifiability of its specific claims.

2.8.3 7.3 Nonlinear Symmetry and Mirror Symmetry

The proposed theory uses nonlinear dynamics to explain how small perturbations can be amplified into a macroscopic chiral bias.

Nevertheless, nonlinear dynamics and mirror symmetry are different concepts.

A nonlinear system can preserve mirror symmetry exactly. Conversely, a system can break mirror symmetry through an external chiral field or asymmetric boundary condition even if its dynamics are not strongly nonlinear.

The relevant question is whether a specified nonlinear reaction network exhibits mirror-equivariant dynamics and, under defined conditions, develops a symmetry-broken state.

The present architecture treats the mirror involution as the mathematical operation used to define the symmetry relation. Nonlinear dynamical selection is the process whose outcomes are investigated.

2.8.4 7.4 Relationship to Existing Chemical Mechanisms

The geometric architecture is intended to complement established reaction-network and kinetic models.

A reaction model may already explain chiral amplification through autocatalysis, mutual inhibition, aggregation, or nonlinear feedback. The geometric programme becomes scientifically useful if it can:

- identify state-space structure not evident from a purely parameter-based description;
- provide a coordinate-consistent geometric representation;
- improve prediction of transition thresholds or robustness;
- connect dynamical structure to experimentally measurable quantities;
- generate predictions that distinguish it from competing explanations.

If the geometric descriptor is merely a re-expression of existing kinetic variables, it may still provide a useful representation but would not establish a new independent physical mechanism.

2.8.5 7.5 Relationship to Geometric Selection Theory

The companion Super Cell paper investigates whether state-space geometry can help characterize cell-fate transitions.

At the molecular scale, the competing states are enantiomeric states. At the cellular scale, the competing states are phenotypic attractors.

The shared architecture is the study of competing dynamical states, their basin boundaries, and possible geometric descriptors of transitions.

However, the molecular and cellular systems differ in their governing equations, observables, timescales, and sources of noise. A common mathematical vocabulary does not establish that they obey identical quantitative laws.

Geometric Selection Theory should therefore be regarded as a cross-domain research programme. Its generality will depend on independent demonstrations that the proposed descriptors are well defined and predictive in each application.

2.8.6 7.6 Implications for the Origin of Life

Homochirality is an important problem in prebiotic chemistry because biological function depends on molecular stereochemistry.

The present architecture suggests one possible line of investigation: determine whether the state-space geometry of prebiotic reaction networks provides predictive information about the emergence and amplification of chiral bias.

This is a conditional proposal. The existence of a geometric descriptor does not imply that every chiral chemical system will spontaneously become homochiral. The outcome depends on the reaction network, thermodynamic constraints, stochastic fluctuations, environmental conditions, and the presence or absence of chiral influences.

A geometric explanation would become compelling if it identified measurable conditions under which chiral amplification becomes more likely, more stable, or quantitatively more predictable than conventional models indicate.

2.8.7 7.7 Limitations

Challenge	Scientific issue	Required next step
Curvature descriptor	The descriptor may depend on orientation, normalization, or model choice	Define it operationally and demonstrate coordinate consistency
Assumption C3	A scalar descriptor need not characterize the full metric symmetry	Prove the equivalence for a restricted model or replace it with a sufficient descriptor
Assumption C4	The coupling between geometry and chiral bias is not universal	Derive it from a specified dynamical model or validate it independently
Nonequilibrium potential	A quasipotential depends on the stochastic dynamics	Specify the stochastic process and action functional
Topological protection	A topological invariant does not automatically protect chemical observables	Identify the invariant, perturbation class, and measurable consequence
Nonlinear threshold	The threshold depends on the chosen control parameter and criterion	Define a dimensionless threshold and derive its scaling
Experimental reconstruction	Finite data may not uniquely identify state-space geometry	Use independent probes, uncertainty quantification, and out-of-sample validation
Cross-scale generalization	Molecular and cellular systems have different dynamics	Validate each application independently

2.8.8 7.8 Future Directions

The immediate research priorities are:

1. **Model-specific proof of C3.** Construct a state-space model in which the curvature descriptor is demonstrably sufficient for the relevant metric symmetry condition.
2. **Independent derivation of C4.** Establish a dynamical relationship between the descriptor and chiral-selection bias without defining one solely through the other.
3. **Reaction-network reconstruction.** Apply the architecture to a specified autocatalytic network and compare its predictions with an established kinetic baseline.
4. **Topological analysis.** Identify a mathematically appropriate invariant and determine whether it predicts persistence of a defined dynamical regime.
5. **Quantitative threshold testing.** Define operational amplification thresholds and compare dimensionless scaling laws.
6. **External-field controls.** Distinguish geometric contributions from established field-dependent reaction mechanisms.
7. **Cross-scale validation.** Investigate whether analogous geometric descriptors are useful in cell-fate models without assuming universal equations.

2.9 8. Conclusion

This work proposes a geometric-dynamical architecture for investigating homochirality through the state-space structure of competing enantiomeric attractor basins.

Its central mathematical result is a conditional equivalence theorem. Under explicit assumptions, a nonzero chiral-selection observable, a nonzero separatrix-curvature asymmetry, and mirror-symmetry breaking of the Fisher metric are equivalent descriptions within the specified model class.

The architecture is grounded in a small set of General Postulates of Geometric Selection Theory (Section 2.1) that define the geometric arena. These postulates play the same logical role as the general postulates of general relativity: they specify the architecture' s objects and their relationships, but they do not by themselves predict any specific numerical outcome.

The manuscript also develops two research hypotheses derived from the postulates. Topological descriptors may contribute to the characterization of autocatalytic robustness when a suitable invariant and its physical significance can be established. Separatrix geometry may influence nonlinear amplification by modifying transition actions or rates, but quantitative threshold laws must be derived from specified reaction dynamics.

Four experimental directions are proposed: testing geometric contributions to prediction, investigating topological robustness, measuring geometric contributions to amplification thresholds, and evaluating geometry-mediated external-field effects.

Together with the companion Super Cell paper, this work advances Geometric Selection Theory as a proposed cross-scale programme for studying whether geometric properties of competing dynamical states can help predict selection outcomes.

The scientific value of the programme will depend on its ability to produce quantitative, falsifiable predictions that improve upon existing explanations. The decisive next step is to reconstruct a specific chemical state space, define its geometric descriptors independently, derive their relationship to the reaction dynamics, and test that relationship on independent experimental data.

If these tests succeed, state-space geometry may become a useful additional principle for understanding chiral selection and nonlinear chemical amplification. Until then, the architecture should be understood as a conditional mathematical architecture and a testable research hypothesis, rather than an established universal law of homochirality.

2.10 Appendix A. Scientific Status of the Main Claims

This manuscript distinguishes three categories of statements.

2.10.1 A.1 Postulates (Architecture-Level)

The following are accepted as architecture-level postulates of Geometric Selection Theory, following the pattern of the general postulates of general relativity:

- **P1.** Statistical state manifold with Fisher information metric.
- **P2.** Mirror involution representing enantiomeric or competing-state exchange.
- **P3.** Nonequilibrium potential with covariant Hessian.
- **P4.** Sufficiently regular separatrix geometry.
- **P5.** Nonzero geometric coupling to chiral observables.

These postulates define the mathematical arena of the theory. They are not individually falsifiable, and they are not empirical claims about any particular reaction.

2.10.2 A.2 Established Mathematical and Scientific Background

The following are established within their respective conditions of applicability:

- Riemannian metrics, connections, curvature, and covariant Hessians are standard mathematical constructions.
- Fisher information geometry provides a canonical metric for regular statistical models under the relevant information-geometric assumptions.
- Freidlin–Wentzell large-deviation theory describes exponential-order behavior of rare transitions for appropriate stochastic dynamical systems.
- Reaction-network models, autocatalysis, and nonlinear symmetry-breaking dynamics provide established approaches to chemical amplification.
- Nontrivial topological invariants can remain unchanged under specified continuous deformations, provided the relevant mathematical conditions hold.

2.10.3 A.3 Proposed Theoretical Architecture

The following are proposals developed in this manuscript:

- A state-space geometric representation of chiral selection.
- The use of a separatrix-curvature descriptor to characterize competing enantiomeric basins.
- The conditional equivalence theorem under Assumptions C1–C4.
- A programme for testing topological descriptors of autocatalytic robustness.
- A geometric contribution to nonlinear amplification through transition-action sensitivity.
- Geometric Selection Theory as a cross-scale research programme.

2.10.4 A.4 Testable Hypotheses

The following remain open research questions:

- Whether C3 holds for any specified chemically realistic model.
- Whether C4 can be derived from a reaction network or independently supported by data.
- Whether geometric descriptors improve prediction of chiral amplification beyond established kinetic variables.
- Whether an appropriate topological invariant predicts a measurable robustness property.
- Whether a geometric threshold law holds for a defined catalyst series.
- Whether external-field effects include an independently measurable geometry-mediated contribution.

No hypothesis in this category is treated as established merely because it is mathematically compatible with the proposed architecture.

A suitable validation programme should include operational definitions, baseline comparisons, uncertainty quantification, independent datasets, and replication.

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2.13 Availability of Data and Materials

Code, mathematical scripts, and preprints are available at GitHub:

https://github.com/Zaitian001/-Self-Preprint-V1.1-

2.14 Competing Interests

Not applicable.

2.15 Author Contributions

Baoliin (Zaitian) Wu 伍宝林 is the sole author of this manuscript and is responsible for all aspects of the research, including conceptualization, methodology, theoretical formulation, and manuscript writing.

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2.17 Physical Proof of Existence (Appendix)

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